

Ray Foysal

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Education

University College Dublin (UCD) | Dublin, Ireland

2023 - Present

- Grade 1:1
- Relevant Modules: Mechanics for Engineers, Mechanics of Fluids, Manufacturing Engineering, Energy Engineering, Physics for Engineers, Computer Science for Engineers, Computer Engineering, Robotics Design Project, Electrical and Electronic Circuits, Creativity in Design.

Colaiste Pobail Setanta | Dublin, Ireland

2018 - 2023

- Leaving Certificate Points Obtained: **613**

Experience

UCD UAV | Aeromechanical & Simulations Engineer

Feb 2024 – Present

- Conducted analysis of air foil designs and lift/drag properties using XFLR5 CFD software.
- Utilized X-Plane 11 simulations to refine flight characteristics of different planes in our prototype series.
- Selected and produced landing gear components based on their strength standards and our requirements. Used SimScale to simulate and verify these properties of the landing gear components.
- Collaborated with the Avionics department to design a transport container for the UAV while innovating by implementing the ground control station to maximise ease of transportation.

UCD Formula Student | Powertrain Engineer

Sep 2023 - Present

- Contributed to the development of the High Voltage Battery Enclosure by designing it with SolidWorks and Onshape.
- Evaluated appropriate high voltage fuses for use in internal battery connections, and created technical reports for review by faculty advisors.
- Involved in the creation of the car's fiberglass hood during production to reduce car's weight by 10%.
- Assembled wheel connections such as tie rods, knuckles, axle, and suspension components.
- Participated in the IMechE FSUK 2024 competition as a member of the UCDFS team and secured 39th position at Silverstone.

Work Experience

Red Cow Moran Hotel | F&B Assistant

Sep 2023 – Present

- Received Manual Handling and HACCP training.
- Worked in a fast paced environment, served 100+ people every day.

Talbot Hotel Stillorgan | F&B Assistant / Receptionist

February – March 2022

- Assisted in daily operations, managed guest inquiries, and helped to ensure a high level of 95% customer satisfaction.

Technical Skills

Modelling/Simulating

- Proficiency in SOLIDWORKS, OnShape and Autodesk Fusion 360
- Knowledge of MATLAB from Engineering Computing course undertaken in UCD.

- Knowledge of Blender 3D modelling software to create several animations for Engineers Without Borders video presentation and for product advertisements for personal e-commerce business.
- Knowledge of Simulating softwares such as SimScale, XFLR5, and X-Plane 11 from experience working on the UCD UAV team.

Programming

- Proficiency of C/C++ Programming from Computer Science, Computer Engineering and Robotics modules undertaken at UCD.
- Knowledge of Python programming language from online courses CS50 Introduction to Programming with Python and Udemy.

Manufacturing

- Knowledge in 3D printing, CNC machining, and manual machining.
- Knowledge of materials and associated properties; thermoset/thermoplastic polymers, composites, metals etc.

Circuitry & Electronics

- Knowledge in LTSpice simulation and KiCAD PCB Modelling.
- Knowledge in Arduino, sensors, shields, breadboard circuitry from multiple projects; Robotics Design Module, MicroMouse Competition, NASA SpaceApps Hackathon, UCD Elecsoc Drone Project.

Microsoft Suite

- Proficiency in MS Excel from specific coursework done in Intro to Eng. Computing, along with MS Word and PowerPoint.

Achievements/Extra Curricular

NASA Space Apps Hackathon | Winner

Oct 2023

TUS Athlone, Ireland

- Achieved first place in the largest hackathon in Ireland, using NASA's open data.
- Responsible for designing and 3d printing the prototype beacon that detects forest fires from its initial stages and warns local authorities.
- Used Onshape CAD software to decrease environmental waste by 25%.

Produced AI Podcast

February 2024

- Served as the producer for a podcast episode on AI as part of a series from the UCD College of Engineering and Architecture.
- Collaborated with the facilitators, team members and guests to plan, script and record the episode.
- Coordinated logistics, including managing recording equipment, booking recording times at a studio and post-production editing to ensure high quality content delivery.

UCD Entrance Scholar

March 2024

- Awarded UCD Entrance Scholar Award for achieving outstanding result in the Leaving Certificate and graduating as the top student in secondary school.

Robotics Design Competition | Software Developer and Design Engineer

May 2024

- Developed and implemented a strategy in C programming for the robot, resulting in a top-half finish in the competition.
- Designed mechanical systems within the robot to combat and outlast opponents.

Colaiste Pobail Setanta | Best Senior Certificate

June 2024

- Awarded to the senior from secondary school with the highest leaving certificate result.

Sports Star of The Year

June 2021

- Recognised as the best performing student athlete in secondary school.